# The Red/Black Game II:

## Solving the Belief-Reset Abstraction Exactly, and Measuring the Distance to the Real Game

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**Abstract**

Paper I formalized heads-up Red/Black and showed that its reachable beliefs factorize over a class prior into two per-seat tilts. This paper solves the abstraction $G'$ that forgets the tilts: Red/Black with beliefs reset to the class prior at every round boundary. The solve decomposes the game into 5,721 subgame-bearing public classes, orders them by a layer index, and runs a vectorized form of CFR+ over each class against solved successors, to per-class exploitability tolerances of $10^{-5}$ to $10^{-4}$; the value of $G'$ is +0.0245 to the opener, a 51.22% win probability, and the solve runs in about two hours on a laptop. The second half measures what the abstraction costs, and returns three results we did not anticipate. First, an earlier pass of the same computation converged by every internal meter while carrying values wrong by up to 0.11, a hundred times its tolerance; the error was invisible to convergence checks by construction and was caught only by a seat-symmetry identity, which the solve of record then enforced exactly. A later audit (Section 6.4) showed that enforcement hid rather than fixed the underlying indeterminacy: the colour-swap automorphism, which the solver never enforces, is violated by the same profile at the same magnitude, because CFR+ selects arbitrarily among equilibrium decompositions and the bootstrap composes that selection noise. A third-generation solve replaces the per-class selection with the unique equilibrium of an entropy-regularized game, annealed toward zero, and passes the colour-swap gate at solve precision on every class, with the regularization's cost stated rather than hidden. Second, the solved profile deliberately loses contests it could contest, flipping its own known off-color card with probability up to 0.97 at some information sets; we prove this action is simultaneously optimal in $G'$ and dominated in the real game $G$, so the divergence between a game and its abstraction is not hypothetical but on-path and measurable. Third, the cost of the belief reset is certified from below: a local best-response probe beats the exactly-solved profile by at least 0.598 per game (of a 2.0 range), and the reset destroys 0.84 of the public posterior in reach-weighted total variation. Solving an abstraction exactly is not solving the game; measuring the difference precisely is what makes the next step, carrying beliefs, well-posed. Every number is regenerated by scripts accompanying the papers.

## 1 Introduction

Heads-up Red/Black, formalized in Paper I, is a two-player zero-sum game of hidden information: a shared 52-card deck of 26 red and 26 black, five cards per seat, and rounds of bid, raise, challenge, and procurement in which the loser of each contest discards a card. Write $G$ for this game. Paper I counted it exactly, at $5.3 \times 10^{34}$ public decision histories and $2.4 \times 10^{33}$ perfect-recall information sets, and proved that at every round boundary the reachable public posterior over the seats' private types factorizes as a strategy-independent class prior $\lambda_\kappa$ times two per-seat tilt vectors (Paper I, Thm. 10.3). This paper solves the abstraction that forgets the tilts.

**Definition 1.1** (The abstraction $G'$; Paper I, Def. 10.6)**.** $G'$ is the game identical to $G$ within each round, except that at every round boundary of public class $\kappa$ the joint law over type pairs is reset to the class prior $\lambda_\kappa$; equivalently, the tilts $g_0$ and $g_1$ of the factorization are replaced by 1.

The abstraction is forced. At $2.4 \times 10^{33}$ information sets there is no regret table for $G$, and any tabular method must summarize. The public class of Paper I summarizes knowledge exactly (it pins the support of the posterior; Paper I, Thm. 10.1) but leaves the shape on that support history-dependent. Resetting to the class prior makes an exact tabular solve possible: since every round removes exactly one card (Paper I, Thm. 4.7), the classes form a layered directed acyclic graph, and $G'$ decomposes into per-class subgames solvable by backward induction over layers.

The first half of the paper carries out that solve. The second half measures what it means, and three of its results surprised us.

1. An internally converged solve of $G'$, green on every convergence meter at tolerance $10^{-3}$, carried values wrong by up to 0.114, a hundred times its tolerance. The error was invisible to round-local convergence checks by construction and was caught only by a seat-symmetry identity (Section 6.2), which the second-generation solve then enforced exactly. Enforcement blinded that meter, and the colour-swap automorphism, which nothing enforces, later read violations of the same magnitude on the enforced solve (Section 6.4); the third-generation solve removes the cause instead of the symptom.

2. The exactly solved profile plays, with probability up to 0.97 at some information sets, an action that flips its own known off-color card and loses a contest it could have contested. We prove the action is simultaneously optimal in $G'$ and dominated in $G$ (Theorem 6.3): the divergence between the game and its abstraction is on-path and measurable, not hypothetical.

3. The abstraction's real-game cost is certified from below. A local best-response probe beats the solved profile by at least 0.598 per game of a 2.0 range (Section 6.6), and the belief reset destroys 0.842 of the public posterior in reach-weighted total variation.

The constructive contributions are the decomposition (boundary placement, private-state structure, row-level accounting), a vector-form CFR+ solver whose second-generation version solves all of $G'$ on a laptop in about two hours, and measurement instruments with explicitly stated scopes, two of which certify statements about $G$ itself. Section 2 imports Paper I's results and states CFR+; Sections 3–7 give the decomposition, the solver, the meters, the results, and what they establish. Heads-up ($N = 2$) throughout, hand size $H = 5$, notation continuous with Paper I.

# 2 Preliminaries

## 2.1 Imports from Paper I

Table 1 lists the results this paper uses without reproof. Boundary conventions: a *boundary* is a procurement resolution, where the contest loser is known but has not yet discarded. A boundary's hand sizes $n_i$ are pre-discard; the round that follows is played with $n'_i = n_i - [\, i = \ell \,]$, where $\ell$ is the seat owing the discard, and $T' = n'_0 + n'_1$ is the *layer index*.

Table 1: Results imported from Paper I.

| Result | Content | Role here |
|---|---|---|
| Acyclicity (Thm. 4.7) | one card leaves play per round; $T'$ strictly decreases | layered DAG; backward induction |
| Public class (Paper I, Def. 6.1) | $(\ell, s_0, s_1)$ with $s_i = (n_i, r^i_{\mathrm{lb}}, b^i_{\mathrm{lb}})$; 6,051 classes at $H = 5$, 330 terminal, 5,485 reachable, 5,721 subgame-bearing | the unit of decomposition |
| Order irrelevance (Paper I, Lem. 5.3) | procurement depends on flip counters, not flip order | procurement lattice |
| Game size (Paper I, Thm. 9.3) | $5.3 \times 10^{34}$ public decision histories; $2.4 \times 10^{33}$ infosets | forces abstraction |
| Support completeness (Thm. 10.1) | the class pins the *support* of the public posterior, not its shape | the reset is an approximation to be measured |
| Rank-one factorization (Thm. 10.3, Cor. 10.4) | reachable posteriors are $\lambda_\kappa \cdot g_0 \cdot g_1$ | defines $G'$; powers the TV meter; Paper III |

### 2.2 CFR+

We use counterfactual regret minimization in its CFR+ variant, and state it as known material. *Regret matching* [1] plays each action of a repeated decision with probability proportional to its clipped cumulative regret, $\sigma^{T+1}(a) \propto [R^T(a)]^+$, uniform when all regrets are non-positive, and guarantees $\max_a R^T(a) = O(\sqrt{T})$. *CFR* [2] extends this to extensive-form games: at each information set $I$ of player $i$ it regret-matches on the *counterfactual* regrets, computed from $v_i(I) = \sum_{h \in I} \pi_{-i}(h)\, u_i(h)$, the expected payoff weighted by opponent-and-chance reach only. Dropping the player's own reach makes the per-infoset regrets decompose: total regret is bounded by their sum, so minimizing locally minimizes globally, and in a two-player zero-sum game a pair of average strategies with average regrets at most $\varepsilon$ is a $2\varepsilon$-equilibrium. *CFR+* [3, 4] accelerates CFR by flooring cumulative regrets at zero after every update, weighting the strategy average linearly in $t$, and alternating player updates; it is the variant that essentially solved heads-up limit hold'em [5]. The output of a run is the *average* strategy; the current strategy need not converge at all, and every value, meter, and exported policy in this paper refers to the average.

*Remark* 2.1 (Three caveats that shape this paper). First, the guarantee is about average regret, not uniqueness: where the equilibrium set is large, two runs can converge to different, equally valid answers, and the type-conditional values this paper bootstraps on are exactly such objects (Remark 3.10). Second, the guarantee is for the game the algorithm actually ran on: we run CFR+ on a decomposition of Red/Black, so small regret bounds exploitability in $G'$, and transferring any claim to $G$ requires a separate measurement. Third, the regret decomposition assumes perfect recall, which holds within a round subgame but fails across boundaries by construction, since a successor is entered through a class that has forgotten the bid history.

## 3 The Decomposition

### 3.1 Subgames and the boundary

**Definition 3.1** (Round subgame)**.** The subgame rooted at a non-terminal class $(\ell, s_0, s_1)$ consists of: the pending discarder's joint (drop, reorder) choice; the other seat's reorder; the bidding phase; and procurement, ending at the resolution *before* the next discard.

The rules order a round reorder $\to$ bid $\to$ procure $\to$ discard, so the segment between two boundaries reads discard $\to$ reorder $\to$ bid $\to$ procure, with the discard first. Terminal classes, whose pending discarder holds a single card, have no subgame: the forced discard eliminates that seat and the value is $\pm 1$. The root is the opposite special case: $\ell = -1$, no discard is pending, the pre-stage is just the two reorders, and the bidding starter is seat 0 by convention rather than $1 - \ell$.

### 3.2 Why the boundary sits at procurement resolution

An earlier design placed the boundary at the end of the round, so subgames ran reorder $\to$ bid $\to$ procure $\to$ discard, matching the rules' own phrasing. That choice is wrong in a way no round-local check could see. The counterexample is small: a seat holding one card, known red, first to act against a seat holding a known red and a known black. By a two-stage matching-pennies argument the true value of this position is exactly 25% to the first seat; the end-of-round decomposition priced it at 15.5%. With the discard at the end of a subgame, the loser's keep-secret decision (which card to keep) falls in one subgame while every response to it falls in the next. The successor, solved standalone against a class prior, committed to near-pure responses conditioned on the opponent's type, and the parent's discard could then choose the card that beats that frozen book. The real game forbids the exploit, because the response cannot condition on a secret it has not seen: the keep-secret and the response are effectively simultaneous. Round-local exploitability was small throughout, and composed best-response brackets confirmed the 15.5% tightly, but within the belief-reset game; only an oracle refusing the reset could see the 9.5-point error.

The fix moves the boundary one discard earlier, so the keep-secret decision and every response to it live inside a single exactly solved subgame. That is Definition 3.1, and it is why the pending discarder's drop and reorder form one joint private choice (Definition 3.4): separating them would let the reorder condition on a drop the opponent never sees, the same leak in miniature. The general lesson is that a decomposition boundary must never cut a coupling between moves the real game makes at the same time. Where it does, each side is solved against a frozen model of the other, and the error can be large while every meter computed inside the decomposition reports success.

### 3.3 Information sets, types, and private states

**Definition 3.2** (Information set)**.** Fix a class $(\ell, s_0, s_1)$ and a seat $i$. An information set of seat $i$ inside that class is a pair (public node, seat $i$'s private state), where the public node is everything both players can see (the class, the bid chain so far, and the flip counters with the colors revealed this contest) and the private state is everything seat $i$ knows that its opponent does not.

**Definition 3.3** (Private type)**.** Seat $i$'s type at a boundary is the pair $(r, x)$: the number of reds in its pre-discard hand, and the number of reds among the cards it has itself discarded. The public bounds truncate the first coordinate to $r^i_{\text{lb}} \leq r \leq n_i - b^i_{\text{lb}}$; the second is invisible to the opponent.

Discarded cards can never be flipped, so $x$ cannot affect any payoff directly, but the deck is finite: a player who has seen $k$ reds among their own five dealt cards should assign an unseen card probability $(26 - k)/47$ of being red, from 0.447 to 0.553 as $k$ runs from 5 to 0. So $x$ is private information about the deck rather than the hand, droppable under an idealized infinite deck and retained because the game is played with 52 cards.

**Definition 3.4** (Private state)**.** For a seat that is not the pending discarder, a private state is an ordering of its $(r,\, n_i - r)$ color multiset together with $x$, one per admissible ordering, of which there are $\binom{n_i}{r}$. For the pending discarder, the drop and the reorder are a single joint private choice: a private state is (drop color × ordering of the surviving $n_i - 1$ cards, $x'$), where $x' = x + 1$ if the dropped card was red and $x' = x$ otherwise. Two paths reaching the same post-discard hand (dropping a red from $r$ reds versus a black from $r - 1$) remain distinct states, because they imply different $x'$ and therefore different beliefs about the deck.

**Example 3.5** (Counting states and rows in one class)**.** Take the class $\ell = 0$, $s_0 = (2, 0, 0)$, $s_1 = (1, 0, 0)$. Seat 0 has made $H - n_0 = 3$ discards, so $x \in \{0, 1, 2, 3\}$ and $r \in \{0, 1, 2\}$; its joint (drop, ordering) choices per $r$ are $1, 2, 1$, so $\text{states}_0 = (1 + 2 + 1) \times 4 = 16$. Seat 1 has 4 discards, $x \in \{0, \ldots, 4\}$, $r \in \{0, 1\}$, one ordering of a single card: $\text{states}_1 = 10$. The in-round sizes are $(1, 1)$: 7 bid nodes (3 acted by seat 1, 4 by seat 0, hence $3 \times 10 + 4 \times 16 = 94$ rows) and 14 procurement nodes ($8 \times 10 + 6 \times 16 = 176$ rows unmasked, of which the consistency mask removes 26, leaving 150). Total: $94 + 150 = 244$ rows.

### 3.4 Decision nodes and rows

Within a round, the public structure is the round automaton of Paper I on the in-round sizes $(n'_0, n'_1)$. Two kinds of node carry decisions: *bid nodes* (color, chain), plus the pre-bid node, $2^{T'+1} - 1$ in all, with the actor alternating along the chain; and *procurement nodes* (color, chain, $x$, $y$), where $x, y$ count cards already flipped off the bidder's and challenger's decks and only the bidder acts. By Paper I's order-irrelevance lemma the procurement nodes form a lattice on the two counters, not a tree of flip sequences. Resolutions are terminal within the round; they carry values, not decisions.

**Definition 3.6** (Row)**.** A row is one (decision node, private state) pair belonging to the seat that acts at that node. Each row holds a cumulative-regret vector and a cumulative-strategy vector, one entry per legal action.

The row count is the size of the regret table and the primary determinant of memory and time. It is not a simple product of nodes and states: different nodes are acted on by different seats, and procurement rows whose flipped prefix could not have matched the bid color are masked out entirely. Table 2 gives the totals by layer; they are structural, derivable without solving anything.

*Remark* 3.7 (A lossless order-class reduction)*.* Those counts enumerate hand orderings in full, though only an ordering's leading run is ever observable (procurement halts at the first off-color card), so the $\binom{r+b}{r}$ orderings of an $(r, b)$ multiset collapse losslessly to $r + b$ order classes. The reduction removes about 12% of private states and, because the saving grows with depth (0% at $T' \leq 4$, 31% at the root), roughly 19% of the total row count: some 62M of 336M, 40M of it at $T' = 9$. The first-generation solver did not take it; the second-generation solver does (Section 4.3).

*Remark* 3.8 (The root is large and cheap)*.* The root has the largest public structure anywhere (2,047 bid nodes, 65,790 procurement nodes) yet only 815,456 rows, about four average layer-9 classes, because nobody has discarded: $x = 0$ for both seats and $\sum_r \binom{5}{r} = 2^5 = 32$ private states

Table 2: Regret-table size by layer, enumerating raw hand orderings (the first-generation layout; Section 4.3 collapses order classes). "Classes" counts those with a real subgame. Layers 8 and 9 hold 81.5% of all rows.

| Layer $T'$ | Classes | Rows | Mean rows/class | Share |
|---|---|---|---|---|
| 2 | 36 | 5,424 | 150 | 0.002% |
| 3 | 132 | 70,056 | 530 | 0.021% |
| 4 | 330 | 546,656 | 1,656 | 0.163% |
| 5 | 686 | 3,181,392 | 4,637 | 0.948% |
| 6 | 1,104 | 13,760,056 | 12,463 | 4.100% |
| 7 | 1,290 | 43,581,784 | 33,784 | 12.985% |
| 8 | 1,260 | 106,941,312 | 84,874 | 31.862% |
| 9 | 882 | 166,740,672 | 189,048 | 49.678% |
| 10 (root) | 1 | 815,456 | 815,456 | 0.243% |
| all | 5,721 | 335,642,808 | | |

each. The public game shrinks with $T'$ but the private state space grows with the number of discards, and the second effect dominates.

### 3.5 Entry beliefs and the reset

A subgame cannot be solved without a distribution over the type pairs it starts from. We use the *bounded hypergeometric* law: treat both hands as drawn from the residual 26/26 budget and renormalize over the region the public bounds allow. This is the class prior $\lambda_\kappa$, and it is the one modeling choice in the design. It differs from the exact posterior: the class determines the support of the posterior exactly (Paper I, Thm. 10.1), but the shape on that support depends on how the opponent has been playing, and cross-round bid history is discarded outright. Solving every subgame against $\lambda_\kappa$ is what makes the solved object $G'$ (Definition 1.1) rather than $G$: CFR+'s guarantees apply to $G'$, and in $G$ the profile is a candidate strategy whose exploitability must be measured. The boundary counterexample of Section 3.2 shows the resulting error need not be small; that is why Section 5 exists.

### 3.6 Layered induction and bootstrapping

Every resolution of a subgame at layer $T'$ identifies a successor class at layer $T' - 1$: apply the realized discard's decay to the entry bounds, raise them to the colors seen this round, and record the new loser. Because $T'$ strictly decreases (Paper I, Thm. 4.7), the classes form a layered DAG and are solved bottom-up, each layer's answers serving as terminal payoffs for the layer above.

**Definition 3.9** (Type-conditional value matrix)**.** Write $m_i$ for the number of seat $i$'s types. The solution of a class returns a matrix $V \in \mathbb{R}^{m_0 \times m_1}$, where $V[t_0, t_1]$ is the seat-0 value when the seats hold types $t_0, t_1$ and both play the solved average strategy.

A parent's resolution looks up its successor class and reads the entry of $V$ matching the types the seats hold at that moment. Only $V$ is needed to solve a parent, so high layers solve against a values-only store rather than full child solutions.

*Remark* 3.10 ($V$ is not unique). Different equilibria of the same subgame can produce different $V$ while agreeing on the aggregate value. Where a private coordinate is payoff-irrelevant but belief-relevant, and $x$ is exactly this, the solver may split a mixed strategy into near-pure branches indexed by that coordinate, and the individual entries of $V$ are then arbitrary across the equilibrium set. The aggregate is pinned; the cells are not. This is a measurement trap, and Sections 4.3 and 5 return to it.

# 4 The Solver

## 4.1 Vector form

Walking the game tree one node at a time, as textbook CFR does, is a poor fit here: a seat's private states are numerous and structurally identical, each facing the same public node with the same legal actions. Instead we hold, at each public node, a matrix whose rows are the acting seat's private states and whose columns are the legal actions, and an iteration becomes a sequence of dense matrix operations over public nodes. Per class, for each seat $i$:

| Object | Shape | Meaning |
|---|---|---|
| `states` | $P_i$ | private states (Definition 3.4) |
| `type_idx` | $P_i$ | type of each private state |
| `regret[node]` | $P_i \times A_{\text{node}}$ | cumulative regret |
| `strat_sum[node]` | $P_i \times A_{\text{node}}$ | cumulative strategy, linearly weighted |
| `reach` | $P_i$ | current reach per private state |
| `entry_wt` | $m_0 \times m_1$ | bounded-hypergeometric prior |

A private state is not compatible with every public node: if a procurement node records three flips all matching the bid color, any state whose top three cards differ is impossible there. Rather than branch, a precomputed boolean mask per (node, seat) zeroes those rows; masked rows are excluded from the row count. The row count of Definition 3.6 is exactly $\sum_{\text{nodes}} P_{\text{actor(node)}}$ under the masks, which is what Example 3.5 computed by hand.

One sweep, for the player being updated: (1) regret-match each node's regret matrix rowwise into a strategy matrix, clipping negatives; (2) propagate reach forward from the entry weights through the pre-stage choice and down the public graph, applying strategies and masks; (3) compute counterfactual values at resolutions by contracting the successor's $V$ against the opponent's reach, and propagate backward, accumulating per-action values; (4) add (action value − node value), weighted by opponent-and-chance reach, into the regret matrix and clip at zero; (5) accumulate $t \cdot \pi_{\text{own}} \cdot \sigma$ into the strategy sum. Then swap players; the average strategy is `strat_sum` normalized rowwise. Step (5) must run at every information set the updating player can reach on their own, whether or not the opponent currently reaches it: gating on opponent reach freezes rows the opponent has temporarily stopped visiting, leaving early garbage in the average and making it spuriously exploitable.

## 4.2 Cost model

Time per class scales as rows × iterations / throughput, with throughput in row-iterations per second. Rows are structural (Table 2); iterations are what convergence demands (Section 6). Layer 9 alone holds half the rows, and layers 8 and 9 together hold 81.5%, so scheduling decisions should be made about those two layers first.

### 4.3 The second-generation solver

The first full solve (Section 6.2) ran the machinery above at tol $= 10^{-3}$ on a cloud fleet. Its autopsy motivated three changes, each a structural observation about the game rather than an optimization trick, that together turned a \$23, twelve-hour fleet job into a two-hour laptop job at $10\times$ to $100\times$ tighter tolerance.

**Ordering collapse.** A seat's private state included its chosen deck ordering, but orderings presenting the same color prefix behavior are strategically identical: the opponent can only ever observe a prefix. Ordering-sensitive states therefore quotient into the order classes of Remark 3.7, the same state-versus-information distinction that collapses procurement states in Paper I, shrinking private state spaces by an order of magnitude at the deep layers.

**Seat-mirror reflection.** A class and its seat-swapped image are the same subgame with the labels exchanged, so only one of each pair is solved and the other is produced by reflection. Mirror antisymmetry, the first solve's sharpest measured failure, then holds exactly by construction, and the work halves. The honest check moves one level up: re-solving a sample of reflected classes directly and comparing prior-weighted aggregates gives residuals of $1.5 \times 10^{-6}$ to $8.9 \times 10^{-6}$, against first-solve mirror gaps of up to 0.114.

*Remark* 4.1 (Equilibrium multiplicity is not error). Cell-wise, a re-solved class and its reflection can disagree by as much as $6.7 \times 10^{-2}$ in individual type-pair conditional values while agreeing to $10^{-6}$ in every prior-weighted aggregate. Zero-sum equilibria pin values, not solutions, and conditional-value cells are selection-dependent (Remark 3.10). Consumers must never compare cells across independently produced solutions, a rule that returns with force in Paper III, where per-type promises are exactly such cells.

**The kernel.** Profiling showed the solve was interpreter-bound: tiny dense matrices under huge node counts. Compiling the sweep, and later the best-response sweeps of Section 5, into a JIT kernel recovered the arithmetic throughput the cost model assumes. With all three changes the full game solves locally in 2h02m at layer tolerances of $10^{-5}$ to $10^{-4}$, and the pipeline is deterministic: re-running it reproduces the solve bit-for-bit.

## 5 Meters

Each instrument below certifies a different claim, and confusing them is the main way to overstate a result. The first four are internal to the decomposition; the last two speak about $G$ (Table 3).

### 5.1 Round-local exploitability

**Definition 5.1** ($\varepsilon_{\text{round}}$)**.** For a single class, hold the solved continuation values fixed and compute both players' exact best responses against the average profile within that round. The sum of their gains is the round-local exploitability.

This certifies that CFR+ has converged on the subgame given its inherited values; computed with successor values held fixed, it is blind by construction to whether those values were right.

## 5.2 Composed exploitability

**Definition 5.2** ($\varepsilon_{\mathrm{comp}}$)**.** Best-respond across the whole class DAG, in the round and in every successor round, against the cached average profile.

This certifies exploitability in $G'$, not in Red/Black. Two sub-meters exist: a *lower bound*, since any concrete deviating strategy's gain bounds the true best response from below, and an *exact* figure for a redeal variant in which types are re-drawn at every boundary, where scalar continuations make backward induction exact. Neither removes the belief reset; they measure inside it.

*Remark* 5.3 (A negative exploitability reading, 2026-07). An early composed-exploitability run reported $\varepsilon_{\mathrm{comp}} < 0$, impossible for a true best response. The cause was the lower-bound structure itself: child-level best response maximizes a child-posterior-weighted total, so a parent can weight exactly the type pairs sitting below the profile's value, and a composition of locally best responses is not a globally best response. The number was a valid lower bound doing what lower bounds do; every meter is now labeled with its bound direction in the code, and an "impossible" reading is treated as a definition surfacing rather than a bug. Separately, a best-response context was found loading per-state tables into per-shared-row storage, misaligning every class where the discarder collapse makes those indexings differ and voiding every composed figure computed before the fix; the kernel best response is now validated against an independent reference implementation on 1,184 classes to $|\Delta| \leq 4.4 \times 10^{-16}$ before any number it produces is reported.

## 5.3 The perfect-recall oracle

**Definition 5.4** (Oracle bracket)**.** Build the explicit perfect-recall game tree over the raw engine, with information sets indexed by a player's full own history and every public observation, and bracket the value with exact best responses.

This certifies the true value in the real game, with no belief reset anywhere, but the tree is explicit, so it is restricted to small classes: it says the most and scales the worst.

## 5.4 The seat mirror

A class and its seat-swapped image must have exactly opposite values under any correct solver: $v(\ell{=}0, A, B) = -v(\ell{=}1, B, A)$. Once classes bootstrap from other classes, mirror discrepancy becomes a cheap proxy for composition error, precisely the quantity $\varepsilon_{\mathrm{round}}$ cannot see. It is an internal instrument both scalable and sensitive to composition, which made it the practical tripwire for the first solve (Section 6.2). The second-generation solve enforces the identity by reflection, which makes this meter read zero by construction; the colour-swap automorphism (Section 6.4) is the surviving measured instrument of the same kind, and nothing in any solver generation enforces it.

## 5.5 Instruments for the real game

The remaining two instruments certify statements about $G$ itself, and both scale. Their target is standard zero-sum exploitability.

**Definition 5.5** (Real-game exploitability)**.** For a profile $\sigma = (\sigma_0, \sigma_1)$ of $G$,

$$\varepsilon_G(\sigma) \;=\; \max_{\sigma_0'} u_0(\sigma_0', \sigma_1) \;+\; \max_{\sigma_1'} u_1(\sigma_0, \sigma_1'),$$

the total gain available to best-responding deviators. It is non-negative, zero exactly at a Nash equilibrium of $G$, and measured per game on the value range $[-1, 1]$ (a 2.0 range).

**Definition 5.6** (TV meter)**.** Play the profile against itself. At every boundary history $h$ with class $\kappa(h)$, compute the exact public posterior $\mu_h$ over type pairs by a forward pass over the factored belief family of Paper I (Thm. 10.3), validated against brute-force replay to $10^{-12}$, and report the reach-weighted total variation

$$\overline{\mathrm{TV}} \;=\; \mathbb{E}_h\Big[\,\tfrac{1}{2}\sum_{t_0,t_1}\big|\mu_h(t_0,t_1)-\lambda_{\kappa(h)}(t_0,t_1)\big|\,\Big],$$

the expectation taken over the boundaries visited in self-play, so each boundary is weighted by the profile's own reach.

The TV meter measures the size of the modeling choice on the profile's own play path, where the choice is being made.

**Definition 5.7** (LBR probe)**.** A local best responder [6] plays one seat against the stored profile. It maintains the exact opponent-type posterior conditioned on the public history and its own private trajectory, prices each of its actions by contracting the profile's per-class continuation values against that posterior, and plays the argmax. Its measured win-rate gain over the profile-vs-profile baseline (mirrored seeds, Wilson interval) is a certified lower bound on $\varepsilon_G(\sigma)$.

The certification comes from the games actually won, not from the derivation, so it holds no matter how crude the responder is. The probe certifies no upper bound: a profile it cannot beat is not thereby safe. Upper bounds are Paper III's subject.

Table 3: What each instrument certifies. No single row is sufficient, and the last two rows are the only ones that speak about $G$.

| Instrument | Scope | Certifies about | Scales? |
|---|---|---|---|
| $\varepsilon_{\text{round}}$ | one class | convergence, given inherited values | yes |
| $\varepsilon_{\text{comp}}$ | class DAG | exploitability in $G'$ (lower bd.) | yes |
| Oracle bracket | one small class | the true value in Red/Black | no |
| Seat mirror | class pair | internal consistency | yes |
| TV meter | play path | size of the belief reset in $G$ | yes |
| LBR probe | whole game | $\varepsilon_G$ lower bound, certified by games | yes |

# 6 Results

The abstraction $G'$ has been solved twice: all 5,721 subgame-bearing classes, first at tol $= 10^{-3}$ on a cloud fleet, then, after the first solve's autopsy, with the second-generation solver. The second solve is the solve of record and is reported first.

## 6.1 The solve of record

The second-generation solver re-solved the entire game in 2h02m on a laptop, eight workers, at layer tolerances of $10^{-5}$ to $10^{-4}$, which is $10\times$ to $100\times$ tighter than the first solve. Root

$\varepsilon_{\text{round}} = 4.9 \times 10^{-5}$; six classes finished above their layer tolerance, all within a $10^{-5}$ band of it. Mirror antisymmetry is exact by construction, and the reflection residual against independently re-solved mirrors is $1.5 \times 10^{-6}$ to $8.9 \times 10^{-6}$. The pipeline is deterministic and reproduces itself bit-for-bit.

$$v(\text{root}) \;=\; +0.02449 \quad\Longrightarrow\quad \mathbf{51.22}\% \text{ to the player who opens.}$$

This is the value of $G'$ of record. The oracle bracket agrees with the solved value to $1.4 \times 10^{-4}$ on every class small enough to bracket ($T' \leq 3$). For scale, the exactly known $T' = 2$ endgame gives its opener $40651/78302 \approx 51.92\%$ (Paper I, Rem. 10.5).

### 6.2 The first solve and its autopsy

The first solve converged by every internal meter. All 5,721 classes met $\varepsilon_{\text{round}} \leq 10^{-3}$ with none hitting the iteration cap (max_iters = 6000); median iterations per layer ran from 100 to 475, U-shaped in depth (275 at $T' = 2$, 100 at $T' = 5$, 350 at $T' = 9$, 475 at the root), so a cost model extrapolated from shallow layers alone underestimates the deep ones. Fully mixed keep-secret classes needed up to $\sim$2,700 iterations against a median of a few hundred; a common default cap of 800 would have silently truncated about 17% of the shallowest layer, and a generous cap is nearly free since the solver exits at tolerance. Layers 2 through 9 ran on a 28-physical-core cloud VM in 43,653 s (12.1 hours) for about \$23; the root, a single unparallelizable class, ran locally in 2,019 s (33.6 minutes), an Apple M-series performance core measuring about 2.2× a cloud core. The first solve priced the root at $v = +0.0266$ (51.33%).

On the round-local meter the solve was uniformly green. The seat-mirror identity, which that meter cannot see, was violated across the 5,720 mirror pairs:

| Layer $T'$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| Worst gap | 0.0002 | 0.0056 | 0.0141 | 0.0288 | 0.1142 | 0.0767 | 0.0495 | **0.1146** |

Median 0.0014; worst 0.1146, over 100× that class's own $\varepsilon_{\text{round}}$ and 5.7 percentage points of win rate. This is the failure Section 5 predicts: $\varepsilon_{\text{round}}$ certifies a class against its inherited values and is blind to whether those values were right. The gap does not decay with depth; it is bimodal, peaking at $T' = 6$ and again at $T' = 9$, and $T' = 9$, the worst layer in the solve, is the layer every game enters on its second round. The identity check was the only tripwire; the solve of record retires the first solve's 51.33%, which was carrying about 0.11 points of composition error, consistent with its own mirror-gap table.

### 6.3 Equilibrium selection, priced

Zero-sum equilibria pin values, not strategies, so within measured indifference there is freedom to select. Two selections were built and priced:

| Profile variant | $G'$ root value | $\varepsilon_{\text{comp}}$ in $G'$ (/game) | suicide rate (raw) |
|---|---|---|---|
| base (solve of record) | +0.02449 | 0.4533 | 3.6% |
| dominance-pruned | +0.02397 | 0.5053 | 0 |
| concealment tie-break | +0.02449 | 0.5003 | 2.05% |

The pruned variant removes the suicide flip (Theorem 6.3) from the action set; it surrenders $5.2 \times 10^{-4}$ of $G'$ root value and +0.052/game of $G'$-composed exploitability, the measured price of refusing an action that is optimal in $G'$ and dominated in $G$. The tie-break selects, among value-equal actions, the one that reveals less; its purity costs $+0.047$/game, roughly $50\times$ the naive per-decision band intuition, because the band bounds the value shift, not the composed-best-response shift. Neither price is visible on any real-game instrument: LBR, raw and guarded arenas, and the bot battery all measure the three variants as identical (Section 7).

### 6.4 The colour-swap audit and the unique-selection re-solve

Red/Black's deck holds exactly 26 red and 26 black cards and no rule distinguishes the colours, so exchanging red and black everywhere is an automorphism of the game and of $G'$ (the class prior is colour-symmetric). It maps a class by swapping each seat's certain-red and certain-black bounds, and a private type $(r, x)$ of a seat with pre-discard hand size $n$ and $d$ discards to $(n - r,\, d - x)$. Because a two-player zero-sum game has a unique value, the value, the per-type conditional values, and (for any single equilibrium) nothing at all is forced about the per-type-pair decomposition; the value and conditionals must agree exactly between a class and its swapped image, whatever equilibrium a solver lands on. No solver generation enforces any of this, which makes it a meter in exactly the sense the enforced seat mirror no longer is.

*Remark* 6.1 (The audit, 2026-08). Measured on the second-generation solve of record (`scripts/check_symmetry.` artifact `symmetry_full.json`; all 2,971 colour orbits), the invariant fails almost everywhere above the leaves: the per-cell decomposition $v_{\text{types}}$ diverges from its swapped image from layer $T'{=}3$ (worst 0.055, rising to 1.67 at $T'{=}8$), per-type conditional values diverge by up to 0.37 ($T'{=}9$), and aggregate class values by up to 0.034, against reported $\varepsilon_{\text{round}}$ of $10^{-5}$ to $10^{-4}$. The profile therefore does not determine the values of $G'$ to better than about 0.03, and its type-conditional advice not to better than about 0.37, no matter what the round-local meters say. The cause is not a colour-dependent defect: re-solving violating orbit pairs against colour-symmetrised leaf values (`scripts/symmetry_discriminate.py`) reproduces both members as exact images of each other to $10^{-16}$, while raw leaves differing by only $10^{-4}$ move cells by 0.14 to 0.45. CFR+ pins the aggregate value but selects arbitrarily among per-type-pair decompositions; the selection is seeded at swap-invariant classes (where one solve on exactly symmetric inputs still selects an asymmetric decomposition, measured at $4.2\times10^{-3}$) and the class-to-parent bootstrap then amplifies it, because the equilibrium correspondence is discontinuous in its leaf inputs. Enforcing the seat mirror in the second-generation solve removed one visible symptom of this indeterminacy and left the indeterminacy itself untouched, as the first-generation autopsy predicted enforcement would.

The third-generation solve removes the freedom instead of a symptom. Each class is solved to the unique equilibrium of the entropy-regularized round game: per information set the policy follows a mirror-descent step $\ell \leftarrow (\ell + \eta\, q/c)/(1 + \eta\tau)$ on logits $\ell$, with $q$ the counterfactual action values, $c$ the row's fixed entry-prior mass, and the temperature $\tau$ annealed down a fixed geometric ladder from 0.3 to $10^{-3}$, last iterate throughout (`scripts/v9_solve.py`; the update is magnetic mirror descent with a uniform magnet under a reweighted dilated entropy, so the regularized equilibrium is unique for every $\tau > 0$). Uniqueness makes the solution equivariant under every automorphism without enforcing any: the colour gate stays live, and the seat mirror could be measured again. The cost is stated rather than hidden: the exact best-response gap of the final policy is floored by $\tau$ (measured $4\times10^{-3}$ at $T'{=}4$ up to 0.19 on the largest $T'{=}9$ class, `scripts/qre_calibrate.py`), while the value bias against the CFR+ value, which is unique and therefore ground truth for values, is $2.7\times10^{-4}$ to $7.8\times10^{-4}$ at $\tau_f = 10^{-3}$ and scales linearly in $\tau_f$. Policies are full support everywhere, so the off-path uniform fallback of Section 6.5 no longer exists, and near-indifferent actions remain visibly mixed instead of being laundered into certainties. The re-solve is deterministic end to end; split across two machines, every one of the 2,288 overlap-checked class values agreed bitwise.

*Remark* 6.2 (What the third generation bought, 2026-08). The re-solve completed on all 5,721 classes. Measured on the colour swap, which nothing in either solver enforces, the worst disagreement anywhere between a class and its swapped image falls from 1.67 to $3.5 \times 10^{-9}$ in the per-type-pair cells, from 0.37 to $3.3 \times 10^{-10}$ in the type-conditional values, and from $3.4 \times 10^{-2}$ to $2.7 \times 10^{-11}$ in the class values. The second-generation profile violates the invariant on 2,206 of 2,971 colour orbits; the third violates it on none beyond a $10^{-9}$ threshold set tighter than the arithmetic.

The strength cost is nil and the strength gain is also nil. Over 1,500 mirrored pairs with play-time guards disabled and no fallbacks on either side, the two profiles are indistinguishable (0.4970, 95% CI $[0.479, 0.515]$, $p = 0.76$). A local best-response probe carrying beliefs to depth 9, run on both profiles with identical deals, returns $\varepsilon_G \geq 0.0780$ $[0.0160, 0.1394]$ for the second generation and $\varepsilon_G \geq 0.1220$ $[0.0601, 0.1829]$ for the third; the difference is not significant ($p = 0.32$) and separating a gap that size would require some 8,100 games per arm. The honest statement is that unique selection buys determinacy and costs nothing measurable, not that it plays better.

One negative result deserves recording because it settles a question three generations of the certificate program could not. Section 1's audit raised the possibility that the leaf-price dishonorability of 0.11 to 0.26 measured per resolve was largely the same defect as the colour inconsistency of 0.11, rather than a limit of depth-limited resolving. Removing the inconsistency entirely does not lower the probe's edge. The limitation is intrinsic.

### 6.5 Playing the profile

The solved profiles were mapped into the game engine and evaluated over mirrored pairs (each deal played from both seats; 2,000 pairs per cell; play-time dominance guards on):

| Profile | vs gen6 CFR | vs heuristic | self-play seat-0 |
|---|---|---|---|
| first solve (historical) | 78.85% [77.6, 80.1] | 87.10% [86.0, 88.1] | 51.0% |
| solve of record | 79.97% [78.7, 81.2] | 87.55% [86.5, 88.5] | 50.4% |
| dominance-pruned | **81.55**% [80.3, 82.7] | 85.25% [84.1, 86.3] | 51.7% |
| concealment tie-break | 80.13% [78.9, 81.3] | 86.88% [85.8, 87.9] | 51.0% |

Across every evaluation the profile drove every decision, with zero fallbacks, zero out-of-range positions, and zero on-demand solves; since a misplaced boundary surfaces as a fallback, this is meaningful evidence that the class mapping is correct, and every joint discard-and-reorder commitment replayed exactly. Self-play seat-0 rates bracket the solved root (51.22%) within their intervals. The deployed profile is the concealment tie-break variant: value-neutral against every measured opponent, strictly more concealing by construction, and it halves the raw suicide mass. The pruned variant trades 2.3 points against the heuristic for 1.6 against gen6; against non-equilibrium opponents, removing a $G'$-optimal action changes matchups in both directions. These win rates do not measure distance from equilibrium, since none of these opponents probes the abstraction; and with guards off, self-play logged suicide flips at 3.6% of procurement decisions and hundreds of impossible bids, so guards-on parity between variants is a floor, not a finding.

## 6.6 The distance to the real game

Everything above is internal to $G'$ or to a bot ladder. The real-game instruments say the following about the solve of record.

**The reset destroys most of the belief.** The TV meter over 2,000 self-play games (11,320 boundaries) reads $\overline{\mathrm{TV}} = \mathbf{0.842}$ [0.839, 0.845], monotone with depth: 0.66 at the first boundary, 0.95 by the endgame layers. The class prior at depth is nearly orthogonal to the belief an exact observer holds. A cross-fitted refinement ladder localizes the loss: replacing the prior with the true class-conditional posterior mean closes the gap only from 0.842 to 0.671, and enriching the class key with freshness bits (distinguishing Paper I's Path A from Path B) adds only about 0.024 more. Most of the destroyed belief is within-class variation that no small public statistic reaches: the tilts of Paper I's factorization, which only actual belief tracking recovers.

**The abstraction's equilibrium is qualitatively wrong in $G$.**

**Theorem 6.3** (The suicide flip is $G'$-optimal and $G$-dominated)**.** *Call a suicide flip a bidder's choice to flip its own deck when its own top card is certainly off-color. In real Red/Black the action is weakly dominated: flipping the challenger's deck preserves the winning branch, loses the same contests, and reveals the opponent's card instead of one's own. Yet in $G'$, at a probe class solved with successor values held fixed, the exact equilibrium plays the suicide flip on path with probability* 0.97*; exact best response prices it* +0.026 *over the dominating alternative; and a solve with the action masked cannot converge under an unmasked best response (its exploitability floors at* 0.115*).*

The mechanism is the belief reset laundering the self-reveal. Busting one's own known card raises one's public bound, and the successor re-derives beliefs from the diffuse class prior, whereas a real opponent's posterior after a bid followed by a deliberate bust would be devastating. The 3.6% on-path suicide rate of the raw profile is therefore not unconverged residue; it is correct play in the wrong game. The second solve, converged two orders of magnitude tighter, left the rate unchanged, retiring the first solve's reading of the phenomenon as unconverged noise: convergence cannot fix an abstraction artifact at any tolerance; only the action set or the beliefs can.

**Certified real-game exploitability.** The LBR probe (Definition 5.7), 3,000 mirrored games per profile, zero fallbacks, certifies

$$\varepsilon_G \;\geq\; 0.598\ [0.569,\ 0.626] \text{ per game (value range } [-1,1]),$$

identically for the solve of record and for every selection variant, against $G'$-composed exploitability differences of 0.05/game between those same variants and per-round convergence of $10^{-5}$.

*Remark* 6.4 (Head-to-head play is not a solve meter, 2026-08)*.* The second solve, converged 100× tighter to $G'$ than the first, lost their raw head-to-head arena: 48.2% [46.8, 49.6], $p = 0.014$, over 4,800 games, a result that would have justified reverting it. A campaign of 216,000 games proved the matchup a dead tie (50.04% [49.8, 50.3]), and deterministic replay reproduced the 48.2% on its exact seed family: an unlucky draw amplified by pair-correlated $p$-values. In $G$ neither profile is an equilibrium, and two non-equilibria are not ordered by their distance to the $G'$ solve; a gate of the form "new solve beats old solve in games" tests pairwise interaction plus sampling variance, not solution quality. The honest instruments are the composed meters and the probe.

## 7 Discussion

**The two axes decouple.** The central empirical finding is that abstraction-internal quality and real-game quality are separate axes, decoupled in both directions: profile variants differing by 0.05/game in $G'$-exploitability measure identically on every real-game instrument, and the profile whose $G'$ meters read $10^{-5}$ to $10^{-4}$ is exploitable for at least 0.598 per game in $G$. A report citing one axis without the other is, on this evidence, misleading regardless of which axis it picks. The defensible claim is the conjunction Section 6 reports: a stated round-local tolerance across all classes, a composed exploitability in $G'$, oracle agreement where computable, mirror consistency, an explicit statement that the profile is an equilibrium of $G'$ rather than of Red/Black, and the real-game instruments alongside.

**What is established, and what is not.** Established: every class of $G'$ is solved reproducibly to its stated round-local tolerance, the oracle agrees where it can see, and the distance from $G'$ to $G$ is measured, with $\overline{\mathrm{TV}} = 0.842$, $\varepsilon_G \geq 0.598$, and one on-path dominated action with its mechanism identified. The colour-swap audit (Section 6.4) bounds what "solved" delivers for the second-generation profile: values of $G'$ determined to about 0.03 and type-conditional values to about 0.37, because CFR+'s equilibrium selection composes as noise through the bootstrap. The third-generation unique-selection solve restores determinacy, with its regularization cost stated alongside. Not established: any upper bound on $\varepsilon_G$, for this or any profile. Every real-game instrument in this paper bounds from below.

**The handoff.** Re-solving boundaries under the exact posterior, the tilts this paper discards, cuts the probe's certified bound from 0.598 to 0.363 at a shallow depth gate, 0.183 deeper, and 0.078 with full carrying, a complete dose-response. But carrying beliefs soundly, and certifying the result against all opponents rather than one probe family, is a different problem with different mathematics. That problem, its theorem, its instruments, and its current honest status are Paper III.

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## A Notation

Continuous with Paper I; new symbols marked †.

| Symbol | Meaning |
|---|---|
| $H$ | hand size at the deal; $H = 5$ |
| $n_i,\ T$ | hand size / cards in play at a boundary (*pre*-discard) |
| $n'_i,\ T'$ | hand size / cards in play during the round; $T'$ is the layer index |
| $\ell$ | seat owing a discard; $-1$ at the root |
| $s_i = (n_i, r^i_{\text{lb}}, b^i_{\text{lb}})$ | seat $i$'s public statistic |
| $(r, x)$ | private type: reds held, reds among own discards |
| $\lambda_\kappa$ | class prior (bounded hypergeometric) of class $\kappa$ |
| $P_i$ † | number of seat $i$'s private states |
| $I$ † | information set |
| $R^T(I, a)$ † | cumulative counterfactual regret |
| $\pi_{\text{own}}, \pi_{\text{opp}}, \pi_{-i}$ † | reach probabilities |
| $v_i(I)$ † | counterfactual value |
| $V$ † | type-conditional value matrix of a solved class |
| $G'$ † | the belief-reset game the solver actually solves |
| $\varepsilon_{\text{round}}, \varepsilon_{\text{comp}}$ † | round-local / composed exploitability |
| $\varepsilon_G$ † | real-game exploitability (Definition 5.5) |
| $\overline{\text{TV}}$ † | reach-weighted total variation (Definition 5.6) |